CO_2 adsorption in a packed bed worksheet

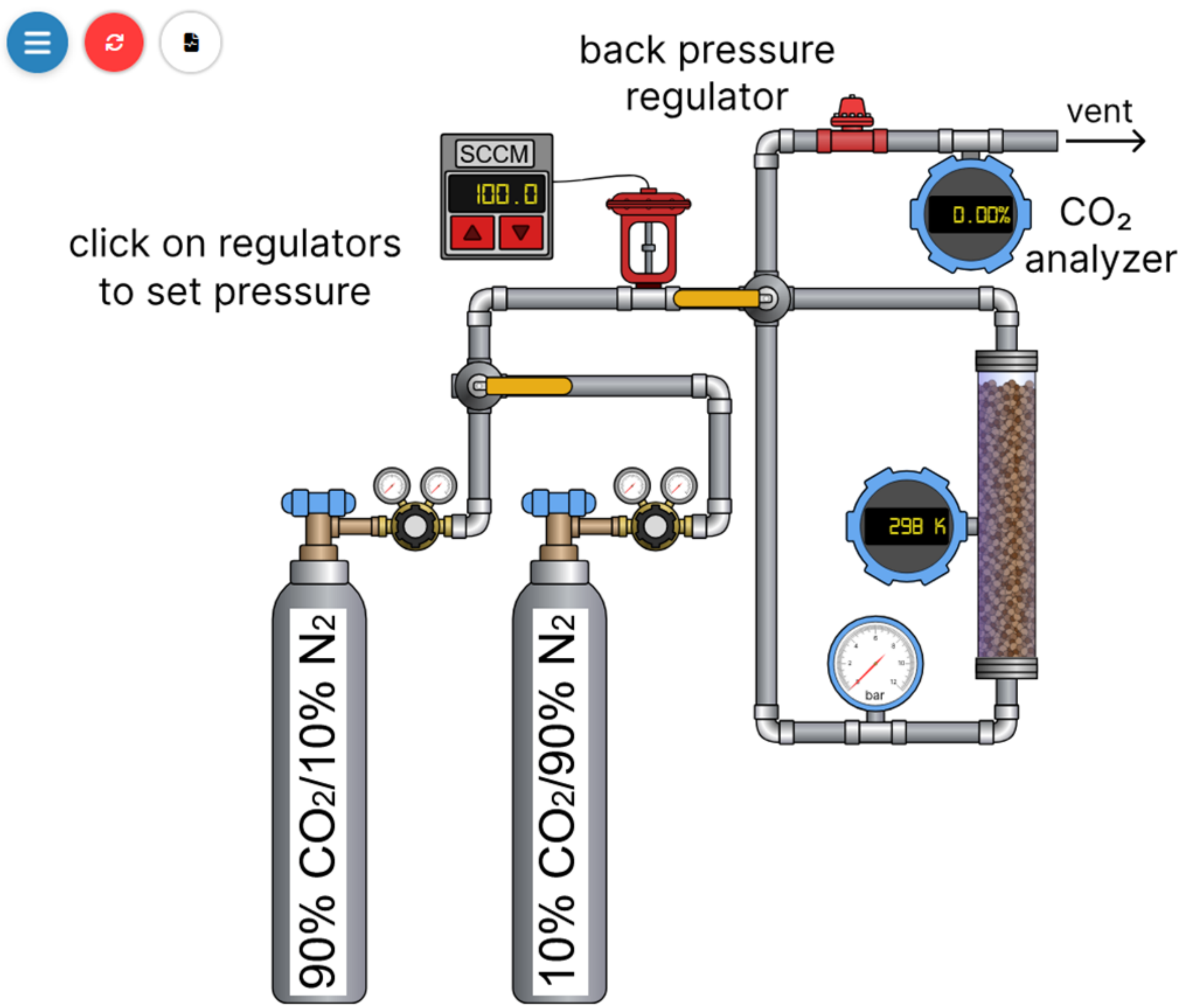
Name(s)____

Carbon dioxide adsorbs on a zeolite in a packed bed column. Breakthrough curves and mass balances are used to measure adsorption capacity, create an adsorption isotherm, and determine the impact of various parameters on adsorption.

Student Learning Objectives

1. Apply mass balances to an unsteady-state flow system.
2. Explain the principles of gas adsorption in a packed bed and the formation of breakthrough curves.
3. Analyze breakthrough data to determine adsorption capacity and interpret the impact of operational parameters.
4. Obtain an adsorption isotherm (amount adsorbed versus CO_2 pressure).

Equipment



- A lab-scale adsorption bed filled with 4 g of zeolite. The bed has a temperature sensor.
- A flow controller that controls and measures the volumetric flow rate into the bed.
- A pressure gauge near bed inlet read gauge pressure.
- A back-pressure regulator to control pressure in the bed.
- Two gas cylinders: 90 mol% CO_2/10% N_2 and 10 mol% CO_2/90% N_2.
- A switching valve selects which gas cylinder flow is fed to the bed.
- A switching valve allows flow to the bottom of the zeolite bed or to bypass the bed and flow directly to the vent.
- A CO_2 gas analyzer to continuously measure the CO_2 concentration at the outlet.

Notation

$y_{CO_2,out}$ = mole fraction CO_2

$y_{CO_2,in}$ = mole fraction CO_2 in the feed to the be

$y_{N_2,in}$= mole fraction of N_2 in the feed to the bed

v = volumetric flow rate into the bed (cm^3/min)

P = system pressure (bar)

P_{CO_2} = partial pressure CO_2 (bar)

R = ideal gas constant ((cm^3 bar)/(mol K))

T = temperature (K)

$F_{CO_2,in}$ = CO_2 molar flow rate into bed (mmol/min)

$F_{CO_2,out}$ = CO_2 molar flow rate out of bed (mmol/min)

$F_{N_2,out}$ = N_2 molar flow rate out of bed (mmol/min) = N_2 molar flow rate into bed

$F_{tot,in}$ = total molar flow rate into bed (mmol/min)

$F_{tot,out}$ = total molar flow rate out of bed (mmol/min)

$n_{CO_2,ads}$ = moles of CO_2 adsorbed

t_{eq} = equilibrium time (min)

MW_{CO_2} = molecular weight CO_2 (44 g/mol)

$m_{CO_2,ads}$ = mass of CO_2 adsorbed (g)

m_{zeo} = mass of zeolite (4.0 g)

q = amount of CO_2 adsorbed per g zeolite (g/(g zeolite))

b = adsorption equilibrium constant (dimensionless)

q_{max} = maximum monolayer CO_2 adsorption capacity (g/(g zeolite))

Questions to answer before starting the experiment

1. If the CO_2 pressure is 10 times as high, do you expect 10 times as much CO_2 to adsorb?

2. How will increasing the total flow rate through the bed affect breakthrough time? Explain

3. How will increasing the CO_2 pressure in the feed affect breakthrough time? Explain

4. The CO_2 concentration at the bed outlet is zero even though CO_2 is entering the bed. Why?

5. Sketch the expected shape of the CO_2 concentration versus time at the bed outlet.

Overview

Measure the amount of CO_2 adsorbed in the zeolite bed by first selecting a volumetric flow rate and system pressure so that sufficient time is available to make measurements. This may require a few runs at different conditions to determine reasonable values of pressure and flow rate. Note that selecting a pressure by adjusting the gas cylinder regulator automatically adjusts the system pressure using the back pressure regulator. Initially CO_2 (either 10 mol% or 90 mol% CO_2) flows to the vent. Click the record data button and then switch flow to the bottom of the bed until the outlet CO_2 concentration equals the inlet concentration.

Experiments

1. Select a cylinder (either 10 mol% or 90 mol% CO_2) and click on the blue valve at the top of the tank to open it. Enter the percent in Table 1. Convert this to a mole fraction $y_{CO_2,in}$ and record that value in Table 1. Then turn the regulator valve clockwise to select a pressure. Zoom with the scroll wheel to make pressure adjustment easier. The pressure is gauge pressure; record that in Table 1. Convert it to absolute pressure and record that value P in Table 1. Calculate the CO_2 partial pressure (absolute) P_{CO_2} in the feed and record the value in Table 1.
2. Select a volumetric flow rate (SCCM, standard cm^3/min) by clicking on the black triangles on the flow controller. Record the volumetric flow rate v in Table 1.

3. If needed, click on the orange valve handle above the cylinders to direct flow from the open cylinder to the vent. When a mouse moves over one of the orange valve handles, the gas flow path displays in green. Lines with no flow are black.
4. Select the record data button (the button to the right of the red reset button) and click on the green start button to start recording data. Then click on the orange handle after the flow controller to change the flow path so gas flows to the bottom of the zeolite bed.
5. The CO_2 analyzer reads zero until breakthrough. Keep recording until the percent CO_2 is equal to the percent CO_2 in the feed cylinder. Then click the red button to stop recording. The data are in a CSV file; download the file by clicking on the button to the right of the stop button. Note that the simulation speeds up time a factor of 60, so one second of simulation time is 1 minute of physical time.
6. Repeat these measurements for a range of CO_2 absolute partial pressures from 1 to 7 bar and record the results in Table 1.

Table 1 CO_2 adsorption on zeolite

	Experiment #						
	1	*2*	3	4	5	6	7
cylinder CO_2 mol%							
$y_{CO_2,in}$							
Gauge pressure (bar)							
P (bar)							
P_{CO_2} (bar)							
v (standard cm^3/min)							
$F_{tot,in}$ (mmol/min)							
$F_{CO_2,in}$ (mmol/min)							
$n_{CO_2,ads}$ (mmol)							
$m_{CO_2,ads}$ (g)							
q (g CO_2/g zeolite)							

Analysis

1. For each experiment, do the following calculations.
2. Calculate the total molar flow rate into the bed $F_{tot,in}$ using the ideal gas law and the volumetric flow rate into the bed v record the value in Table 1.

   $$F_{tot,in} = \frac{Pv}{RT}$$

3. Calculate the CO_2 molar flow rate into the bed $F_{CO_2,in}$ and record the value in Table 1.

   $$F_{CO_2,in} = y_{co_2,in} F_{tot,in}$$

4. Since the CO_2 molar flow rate out of the bed $F_{CO_2,out}$ changes with time because CO_2 adsorbs, calculate $F_{CO_2,out}$ from the total molar flow rate out $F_{tot,out}$ and the fact that the N_2 flow rate does not change.

   $$F_{CO_2,out} = y_{CO_2,out} F_{tot,out} = y_{CO_2,out}\left(F_{CO_2,out} + F_{N_2,out}\right)$$

   and since N_2 does not adsorb, $F_{N_2,out} = F_{N_2,in} = y_{N_2,in} F_{tot,in} = (1 - y_{CO_2,in})\, F_{tot,in}$

   therefore, $F_{CO_2,out} = y_{CO_2,out}\left(F_{CO_2,out} + (1 - y_{CO_2,in})\, F_{tot,in}\right)$

   rearranging, $(1 - y_{CO_2,out}) F_{CO_2,out} = y_{CO_2,out}(1 - y_{CO_2,in})\, F_{tot,in}$

   $$F_{CO_2,out} = \frac{y_{CO_2,out}\left(1 - y_{CO_2,in}\right) F_{tot,in}}{(1 - y_{CO_2,out})}$$

   Thus, calculate the CO_2 molar flow rate out $F_{CO_2,out}$ as a function of time from the values fixed at the start of the experiment ($y_{CO_2,in}$ and $F_{tot,in}$) and the measured CO_2 mole fraction out $y_{CO_2,out}$, which is a function of time.

5. Open the downloaded CSV file and save it as a spreadsheet file.
6. Create a column in the spreadsheet to calculate $F_{CO_2,out}$.
7. Create another column for $(F_{CO_2,in} - F_{CO_2,out})$, which is the difference between the inlet and outlet CO_2 mole fractions.
8. Use the unsteady-state mass balance on CO_2 around the zeolite bed:

   accumulation = in - out

   $$n_{CO_2,ads} = \int_0^{t_{eq}} (F_{CO_2,in} - F_{CO_2,out}) dt$$

   where t_{eq} is the time where equilibrium is reached ($F_{CO_2,in} - F_{CO_2,out} = 0$)

9. Integrate $(F_{CO_2,in} - F_{CO_2,out})$ with respect to time in the spreadsheet to obtain the area under the curve in Figure 1. The area (the shaded rectangle plus the cross-hatched area combined) is equal to the moles of CO_2 adsorbed. Record this value in Table 1.
10. Use the molecular weight of CO_2 to calculate the mass of CO_2 adsorbed $m_{CO_2,ads}$ and record this value in Table 1.

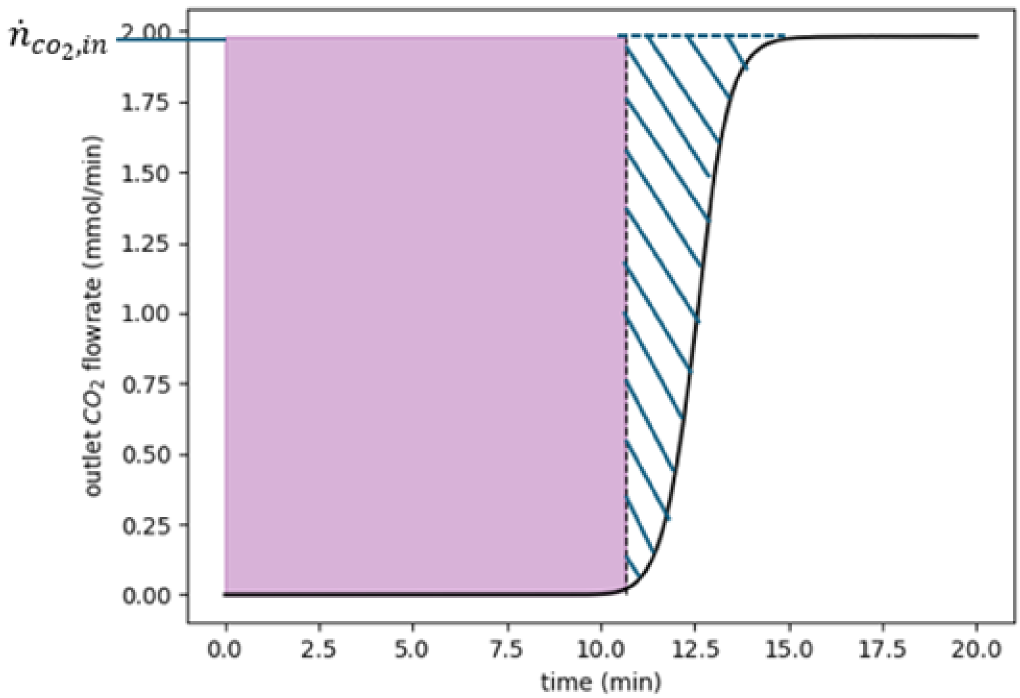


Figure 1 CO_2 outlet molar flow rate versus time for CO_2 adsorption

11. Calculate the adsorption capacities ($q = \frac{m_{CO_2,ads}}{m_{zeo}}$) and record the value in Table 1.

12. After the calculations from all the experiments are in Table 1, plot the amount adsorbed versus CO_2 pressure to obtain an adsorption isotherm.

13. Fit the isotherm data to a Langmuir isotherm and determine the constants q_{max} (maximum monolayer CO_2 adsorption capacity) and b (adsorption equilibrium constant).

$$q = \frac{q_{max} b P_{CO\,2}}{1 + b P_{CO_2}}$$

Questions to answer

1. If two experiments have the same CO_2 partial pressure but different total pressures and feed compositions, do they give the same amount adsorbed? Should they?

2. The temperature of the zeolite bed is constant during this experiment. Would the temperature increase or decrease if the bed were adiabatic? Explain

3. Where might these measurements have errors?

4. How would you use a packed bed on a large scale to trap and remove CO_2 from a stream? Why use multiple beds for continuous operation?

5. What safety precautions would you take to conduct this experiment in the laboratory?